

Architectural Particular Specification
FALL ARREST SYSTEM
SECTION 11.4

Rev	Date	Description
0	14 APR 2022	First Issue
A	13 MAY 2022	Tender Addendum No. 2

SECTION 11.4

FALL ARREST SYSTEM

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. To design, supply and install fall arrest system(s) and all associated fixing and accessories for areas as indicated in Drawings, and shall include necessary associated personnel protective equipments.
- B. Contractor shall submit detail shop drawing, provide mock up, and be responsible for arranging all necessary testing and submitting test reports, as well as to provide operation, maintenance, and safety manual.
- C. Contractor shall design and coordinate the fall arrest system(s) such that the respective structural integrity and watertightness of the roofs to which the fall arrest system(s) are installed will not be affected. Reference

1.02 QUALITY ASSURANCE

(NOT USED)

1.03 PERFORMANCE REQUIREMENTS

- A. The fall arrest system manufacturer should have quality system comply with ISO9001 quality standard. The standards and Codes of Practice specified in this section are listed below for information only, the Contractor shall comply with the edition of the standard or Code of Practice listed in the text when stated or the latest edition when no edition is listed.
 - 1. EN 795
 - 2. EN 360
 - 3. EN 361
 - 4. EN 355
 - 5. EN 353-2
- B. Fall arrest system to Class C per EN795 and shall comprise of cable and anchor bracket system and shall be supplied by an approved proprietary supplier.
- C. The system shall conform to all Buildings Department and other government requirements. The system shall offer hands free use, whilst providing continuous attachment avoiding the need to disconnect and reconnect at anchor points, i.e., allow users to walk uninterrupted the entire length of the system without having to unhook from the system to pass through intermediate support points.
- D. The loads generated by a fall shall consider free fall distance. System layout, anchor system and type of system shall be designed to minimize the effect of the fall on the worker. The system shall be designed to support 4 users MINIMUM in case of a fall and to prevent the users from free falling. Ensure any anchorage used in a fall arrest system is capable of successfully transferring the loads to a level where the building fabric is capable of withstanding them without penetrating the weatherproof layer.

- E. If individual wall mounted anchor points are provided for fall restraint lifeline, the anchor points shall be able to control or restrict the travel distance of a person wearing a full body harness attached to anchor point from reaching the nearest edge of falling hazard. Each anchor point shall be able to support at least 600kg (6kN) per person per anchor point and shall be able to cater for minimum 3 persons. Fixed length lanyard with energy absorber / damper and extended wires shall be provided.
- F. All components shall be designed by the fall arrest system supplier accredited with ISO9001 and shall meet the applicable requirements of EN 795. Fall arrest system shall consist of safety cable attached to the nearest permanent structure.

1.04 ENVIRONMENTAL REQUIREMENTS

(NOT USED)

1.05 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.06 SUBMITTALS

- A. Proposed works programme shall be coordinated and match with contractor's master programme.
- B. In planning stage, the Contractor shall allow sufficient time for the Architect's review and for re-submission by the Contractor where necessary, prior to the commencement of fabrication or works on site. When requested by the Architect, the Contractor shall submit additional items of samples and drawings for review.
- C. Upon award of the Contract, the Contractor shall submit to the Architect for review a checklist of proposed samples and shop drawings, and their intended submission dates. This shall include, but not limited to the items and drawings listed in this Section.
- D. Shop Drawings indicating the following particulars shall be submitted to the Architect for review without objection:-
 - 1. Indicate profiles, sizes, connection attachments, anchorage bolts and type of fasteners and accessories. Include erection drawings with elevations and details where applicable.
 - 2. Layout and elevation drawings shall be drawn no smaller than 1 in 20. Detail drawings at no smaller than 1 in 1.
 - 3. Full extent of fall arrest systems to be clearly described and detailed in shop drawings.
- E. Provide Instructions for the Employer as A4 or A3 sized bound documents in English and Chinese, identified with building name, address, and the date the document was prepared.
- F. Operating Procedures Manual: Provide all necessary information in written form with illustrations, to instruct the Employer's maintenance personnel in the safe use of fall protection systems equipment. Include at least the followings:-
 - 1. Legible roof plans with as-built schematic of installed fall protection systems for each roof. Include manufacturer's serial number, name and part number of each individual component used in the systems.
 - 2. Identification of all anchorage points for fall protection systems;
 - 3. Identification of all anchorage points accessible for decorative fins maintenance;

4. Identification of all personal fall protection requirements and procedures for using equipment;
5. Identification of all dangerous areas on the roof by highlighting all of the "Danger Zone" on the roof plans;
6. Identification of equipment limitations, load ratings, and special use conditions;
7. Provisions for pre-operational, operational, and maintenance inspections;
8. Identification of the access to and egress from work locations and storage areas for fall protection systems equipment;
9. Emergency and rescue procedures, and means of communications to be used during such procedures; and
10. Methods to be used to control employee exposure to falls while they are in the "Danger Zone".

G. Equipment Manual & Inspection Log Book

Manufacturer's standard documentation with "Initial Inspection - Certification for Use" and "Inspection Sign-Off" forms completed. Include (without limitation) contact information for equipment manufacturer's technical support services, and for warrantee matters.

H. As-Built Drawings

A3 sized, plastic laminated, as-built drawings showing fall protection systems equipment locations and details

1.07 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.08 DELIVERY STORAGE AND HANDLING

Package and store materials in a manner that will prevent damage, distortion, breakage or structural weakening. Replace any materials damaged during manufacture, shipping, storage, or installation.

1.09 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.10 TESTING

Provide 100% site test for all fall arrest anchors

1.11 WARRANTY

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

Warranties and guarantees specified in this Article shall not deprive the Employer of other rights the Employer may have under other provisions of the Contract Documents and are in addition to and run concurrent with other warranties and guarantees made by the Contractor under requirements of the Contract Documents

The Management Contractor and qualified applicator shall jointly provide a **ten (10) year** warranty against material defects and workmanship for Work of this Section

Material defects shall include, but not be limited to, chalking, corrosion, discoloration, add adhesion or structural and material failure, ultraviolet and ozone degradation during the warranty period

1.12 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

(NOT USED)

2.02 MATERIALS / SYSTEMS

- A. All materials shall be new and the complete fall arrest system, except for accessory equipment, shall be the product of one manufacturer regularly engaged in the production of such equipment.
- B. The entire horizontal fall arrest system shall be constructed of type 316L (or higher) stainless steel. No polyamide plastic component is allowed for any part of the entire system.
- C. Supporting structure or post shall be galvanized steel with Fasteners.
- D. The Fall Arrest System shall be attached to the supporting structure with appropriate fasteners designed to support a load on the fall arrest system of twice the maximum design load without failure.
- E. Where through bolting is not possible/practical/desirable, reliable proprietary structural fastening systems might be acceptable subject to approval by Architect.
- F. System Components:-
 - 1. Cable shall be continuous or shall have swaged splices, which allow the user to pass without unhooking from the system. The system shall include line tension device, tension indicator, and end brackets (all three of stainless steel) to attach the end of the cable to the structure. End of stainless steel cable should be swage by swaging machine to avoid any risks of incorrect assembly or inadvertent disassembly. End of cable should be swage by swaging machine to avoid any risks of incorrect assembly or inadvertent disassembly. Swage-less type tension device is not acceptable.
 - 2. Support cable not to exceed 5m maximum intervals between type 316L stainless steel intermediate brackets designed to allow user to pass without unhooking from the cable. Brackets to incorporate an energy dissipation system to assist in reducing the fall arrest forces should a user take a fall. All intermediate brackets should be identified by a batch. Hanger that does not hold the cable in upright position with minimum clearance from the supporting fixture is not acceptable.
 - 3. Provided one 316L stainless steel pass through "shuttle" for each user of the system. Shuttle to be provided with double locking carabiner. The shutter shall be capable of being connected or disconnected at any point on the cable and pass over intermediate brackets without having to be detached.
 - 4. Corner (where applicable): 316L stainless steel corner tube section to facilitate making the transition.
 - 5. Brackets and supports should be attached to the structure with appropriate anchors of proper size and embedment to adequately support the intended load.
 - 6. Intermediate bracket must be able to be easily replaced in case of person fall.

G. Lanyard

Provide shock-absorbing lanyard (to EN 355), retractable lifeline or guided type fall arrest rope where applicable for each user of the system.

H. Support Harness

Full-body harness to EN361 with back “D” ring per each user of the system shall be provided.

I. Provide at least 4 sets of 316L stainless steel ‘shuttle’ together with the Lanyard and support harness for the entire project. All the above equipment shall comply with corresponding European Standard.

J. The overall weight of the system shall not exceed the allowance made in the structural design of the building by the project structural engineering consultant.

2.03 ANCILLARY MATERIALS

(NOT USED)

2.04 FABRICATION

A. Assemble, fabricate and install to Manufacturers recommendation and specification.

1. System components shall all be of the same material unless otherwise indicated.
2. Exposed work shall be true to line and level with accurate angles, surfaces and with straight square edges.
3. Coordinate anchorage system with supporting structure. Fabricate and locate anchoring devices as recommended by manufacturer to provide adequate support for intended use.
4. Fabricate joints in a manner to discourage water accumulation. Provided weep holes to drain any water, which could accumulate in the exposed joints.

2.05 FINISHES/FINISHING TREATMENTS

(NOT USED)

2.06 STORAGE

(NOT USED)

2.07 WORKMANSHIP

(NOT USED)

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

- A. After the safety system is installed and properly tensioned, the safety system manufacturer's approved authorized representative shall inspect and operate the system and shall make all final adjustments for proper operation. Each individual end bracket must be load tested and certified safe for use. Report of the inspection shall be submitted to Architect within one week.
- B. Employer /Operator Training
 - 1. Provide the Employer/operator walk through of system and training at the end of the installation procedures. Training is to take the form of a single class conducted on the last day of the installation immediately following completion of the lifeline installation.
 - 2. Engage a manufacturer-authorized service representative to train the Employer's maintenance personnel to operate, adjust, and maintain equipment specified in this Section.
 - 3. Check each fall protection system with the Employer's maintenance personnel present within one month before date of Practical Completion. Determine that operation, systems and devices are functioning properly. Repair all malfunctions..

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION / APPLICATION

- A. The whole fall arrest system should be designed, supplied and installed by trained, approved and competent specialist as appointed by manufacturer. Valid training certificate and manufacturer's company letter should be provided for review prior to installation. The installer must provide evidence of at least 10 similar projects in size and complexity, which have been performed within the previous year.
- B. Install in accordance with approved shop drawings and manufacturer's instructions.
- C. Install anchorage and fasteners in accordance with manufacturer's recommendations to obtain the allowable working loads listed on the engineering data sheet and in accordance with this specification.
- D. Do not load stresses onto the fall arrest system until all materials and fasteners are properly installed and ready for service.
- E. Anchorage Drilled into Concrete
 - 1. Follow manufacturer's recommendations.
 - 2. Do not cut or damage reinforcement
 - 3. Clean drill hole to remove debris and contaminants with manufacturer's instructions.
 - 4. Install anchorage at indicated locations, clearance and embedment depth.
 - 5. When clearance and spacing of anchorage is not indicated on drawings, install at location recommended by manufacturer to obtain maximum working loads in accordance with the distances provided on the engineering data sheet.

- F. Check all units in accordance with manufacturer's instruction, coordinate with adjacent construction for anchor placement, check members true to line, plumb, and level as appropriate.

3.04 TOLERANCES
(NOT USED)

3.05 CLEANING
(NOT USED)

3.06 PROTECTION
(NOT USED)

END OF SECTION